

VGG, ResNet, and Skip Connections

Deep Learning

Daniel E. Acuna

Associate Professor, University of Colorado Boulder

Contents of This Video

In this video, we will cover:

- The trend toward deeper networks
- VGG: simple, uniform architecture
- The degradation problem in deep networks
- ResNet and skip connections
- Why skip connections work

The Depth Trend

Year	Network	Depth	ImageNet Top-5 Error
2012	AlexNet	8 layers	16.4%
2014	VGG-16	16 layers	7.3%
2014	VGG-19	19 layers	7.3%
2015	ResNet-152	152 layers	3.6%

Deeper = Better? Not always—there's a catch...

VGG: Simple and Uniform

Key insight: Use only 3×3 convolutions everywhere

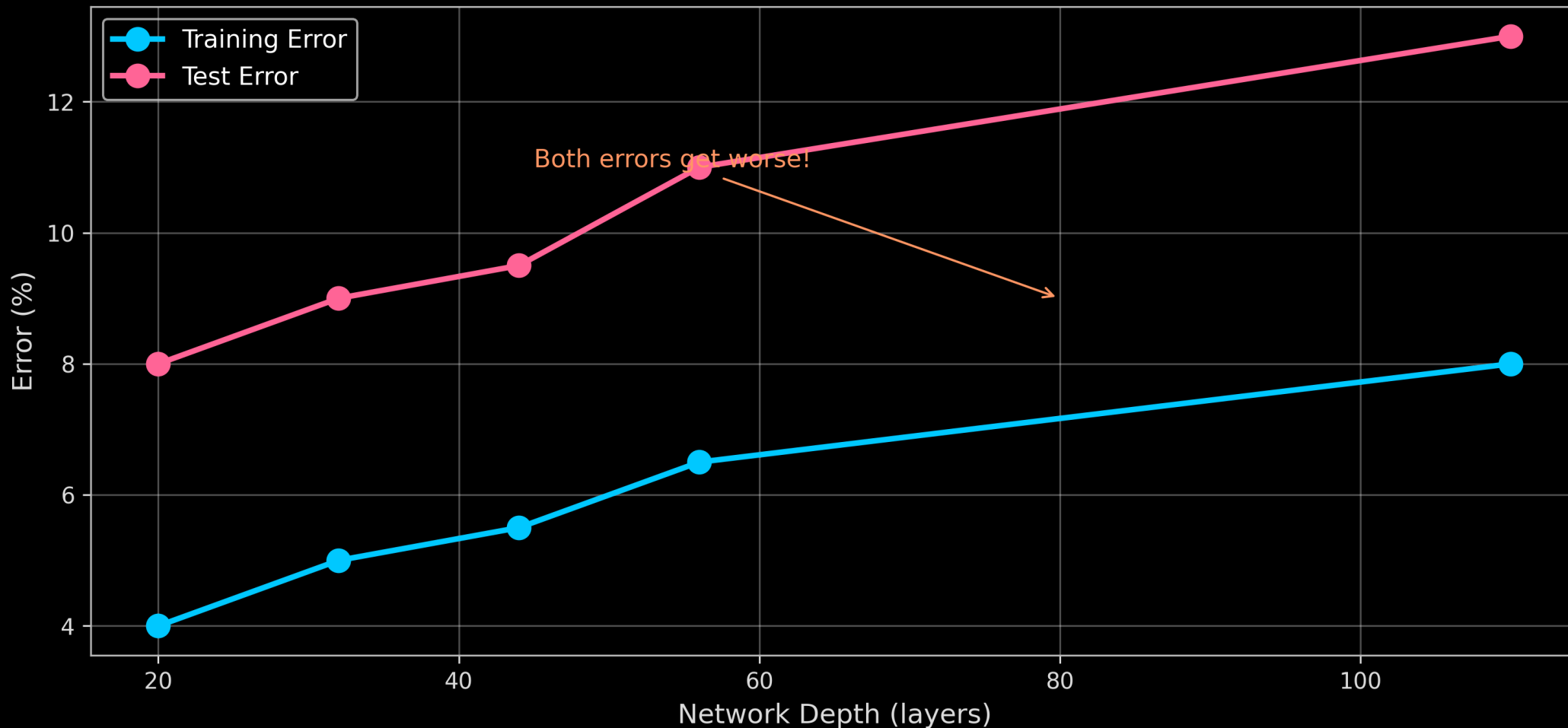
VGG-16 structure:

- 2 conv blocks: 64 filters
- 2 conv blocks: 128 filters
- 3 conv blocks: 256 filters
- 3 conv blocks: 512 filters
- 3 conv blocks: 512 filters
- 3 FC layers

~138 million parameters (large!)

The Degradation Problem

The Degradation Problem: Deeper $\neq$ Better

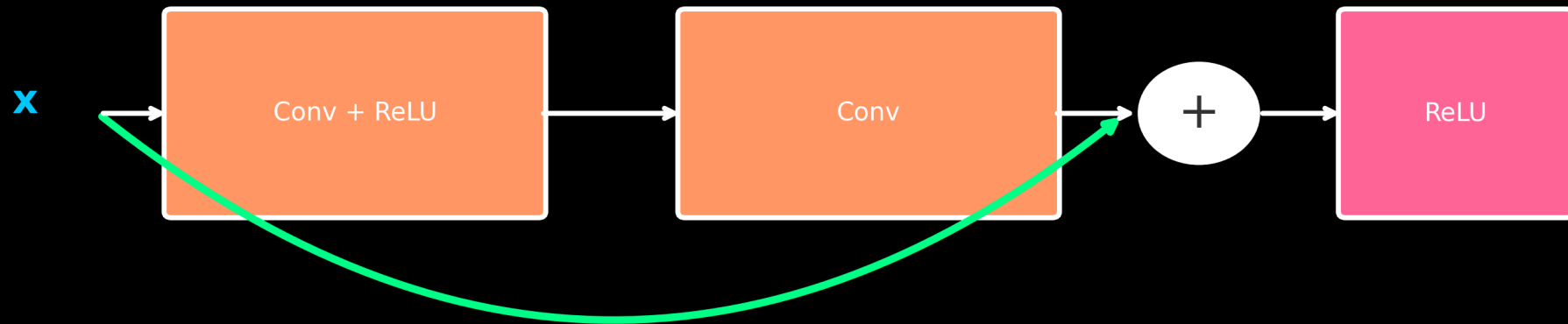


This is NOT overfitting! Training error increases too.

ResNet: The Skip Connection Solution

ResNet Block: Learning Residuals

Skip Connection (Identity)



$$y = F(x) + x$$

Output = Residual + Input

Why Skip Connections Work

Problem: Learning identity mapping is hard for conv layers

Solution: Make identity the default!

$$\mathbf{y} = F(\mathbf{x}) + \mathbf{x}$$

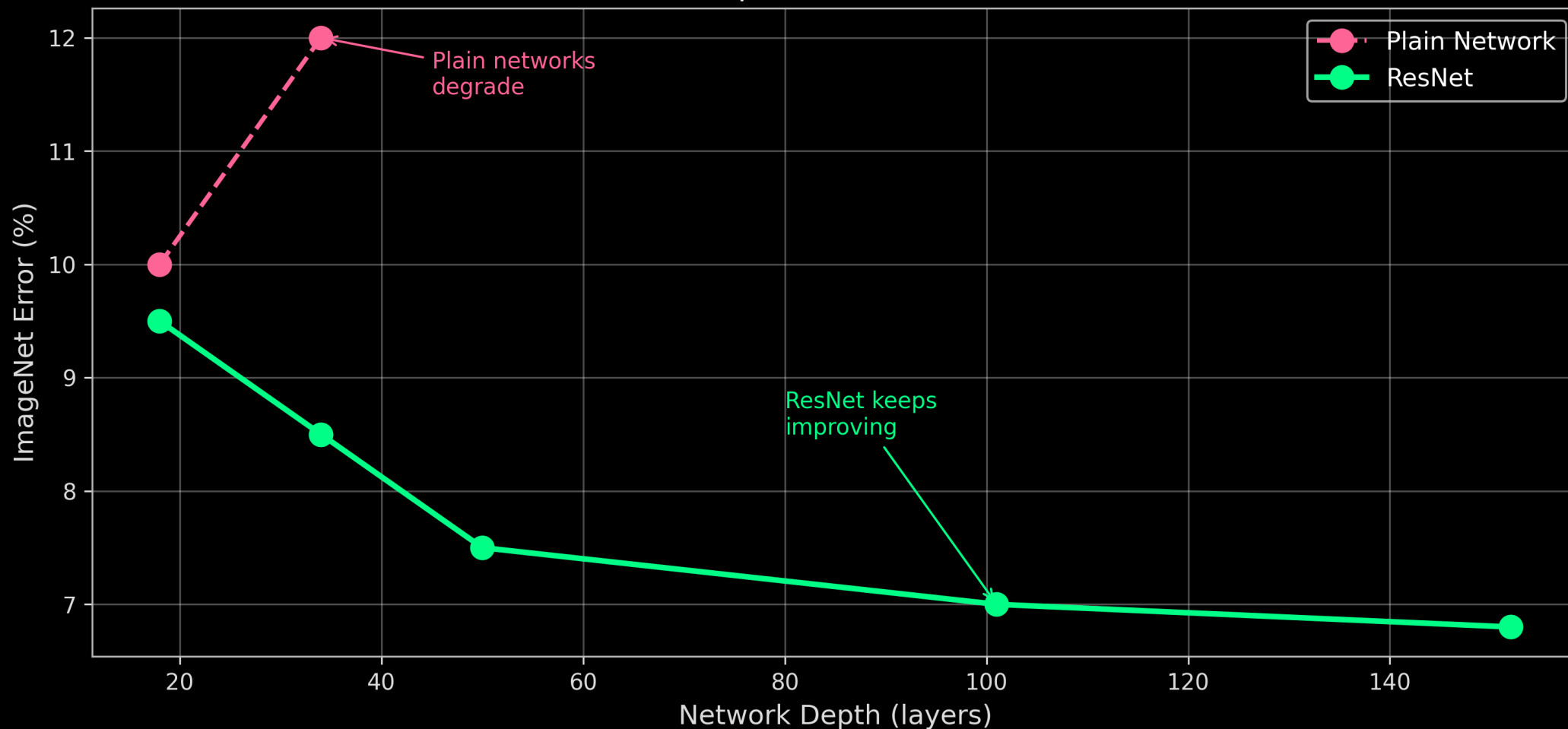
If optimal output $\approx$ input:

- Without skip: must learn $F(\mathbf{x}) \approx \mathbf{x}$ (hard)
- With skip: must learn $F(\mathbf{x}) \approx 0$ (easy, just push weights to zero)

Gradient flow: Skip connections provide direct paths for gradients

ResNet Results

ResNet: Deeper Networks Now Work!



Modern Architectures Build on These Ideas

ResNet variants:

- ResNeXt, Wide ResNet, SE-ResNet

Other architectures with skip connections:

- DenseNet: Connect every layer to every other layer
- EfficientNet: Balanced depth/width/resolution
- Vision Transformer (ViT): Skip connections in attention

Key takeaway: Skip connections enable very deep networks

What We've Covered

In this video, we've learned:

- VGG: simple, uniform architecture using only 3×3 convs
- Degradation problem: very deep plain networks train poorly
- ResNet: skip connections that learn residuals $y = F(x) + x$
- Skip connections make identity mapping easy (learn $F(x) \approx 0$)
- Skip connections enable training 100+ layer networks
- Modern architectures universally use skip connections